

Abstract

Integrating machine vision with autonomous control in smart greenhouses is essential yet challenging. The direct application of deep neural networks in physical control loops often causes instability due to environmental noise and the "black-box" nature of these models. To overcome this, this research proposes a novel "Cyber-Physical Digital Twin" architecture for the real-time monitoring and autonomous management of plant environments. The perception layer utilizes a U-Net model for 2D canopy segmentation (via overhead images) and an EfficientNetB0 network for microscopic disease detection (via macro image sequences). To stabilize this layer, a 1D discrete Kalman filter is deployed in the processing core. By filtering visual noise and enforcing kinematic constraints, it successfully extracts the hidden "growth velocity" variable and guarantees control loop stability even during temporary sensor failures. In the decision-making layer, a rule-based controller leverages the filtered data to autonomously adjust environmental actuators (temperature and electrical conductivity) in response to growth stagnation or biological stresses. Furthermore, to ensure system transparency, an Explainable AI (XAI) dashboard featuring Uncertainty Edge Maps and Grad-CAM was implemented, enabling human operators to visually audit the network's diagnostic decisions. Simulation and stress test results demonstrate that integrating deep learning with kinematic filtration significantly enhances the reliability of agricultural automation, providing a robust foundation for the Industrial Internet of Things (IIoT) in precision agriculture.

Key Words:

Digital Twin, Cyber-Physical Systems, Closed-loop Control, Deep Learning, Kalman Filter, Explainable AI (XAI), Precision Agriculture.